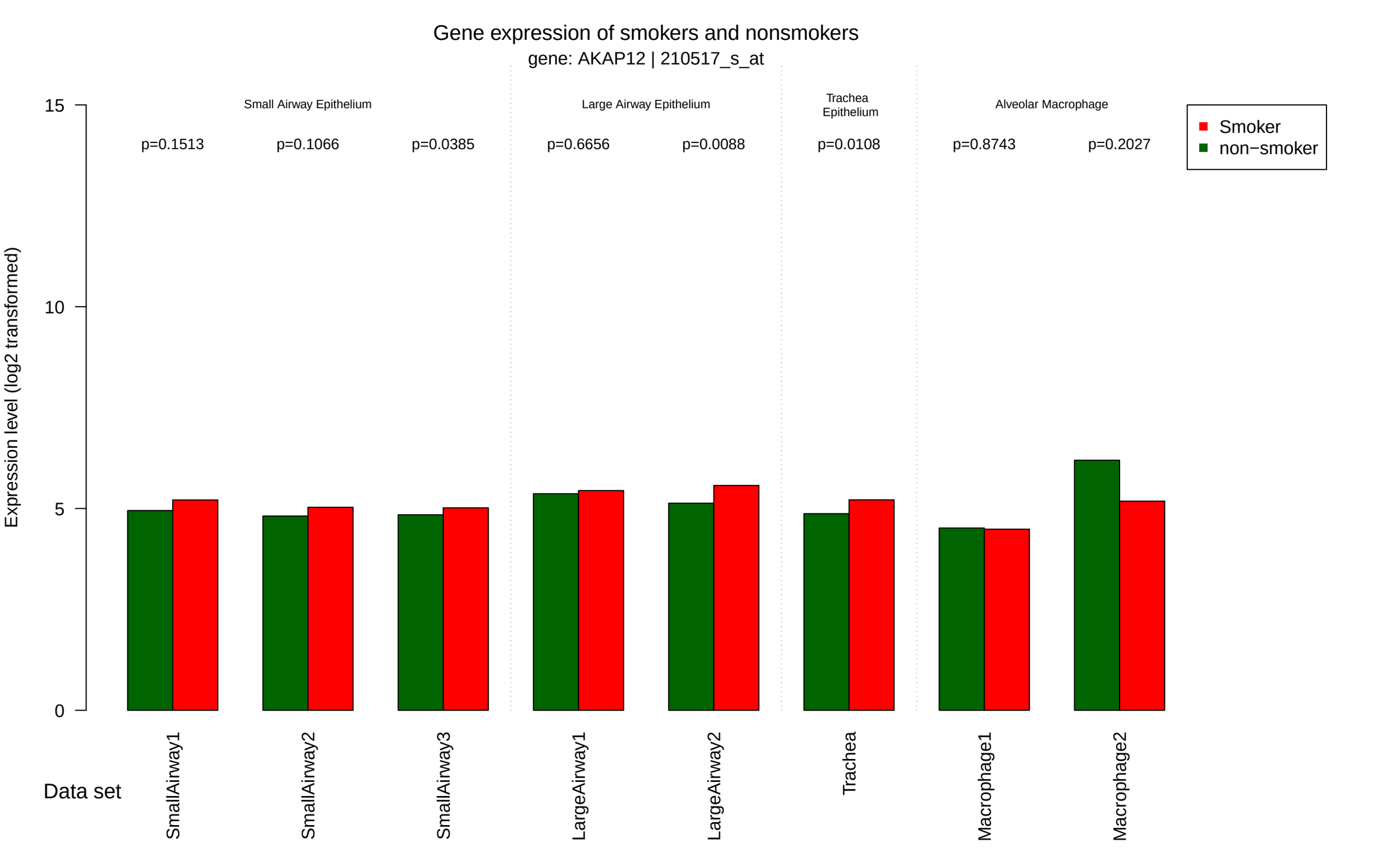
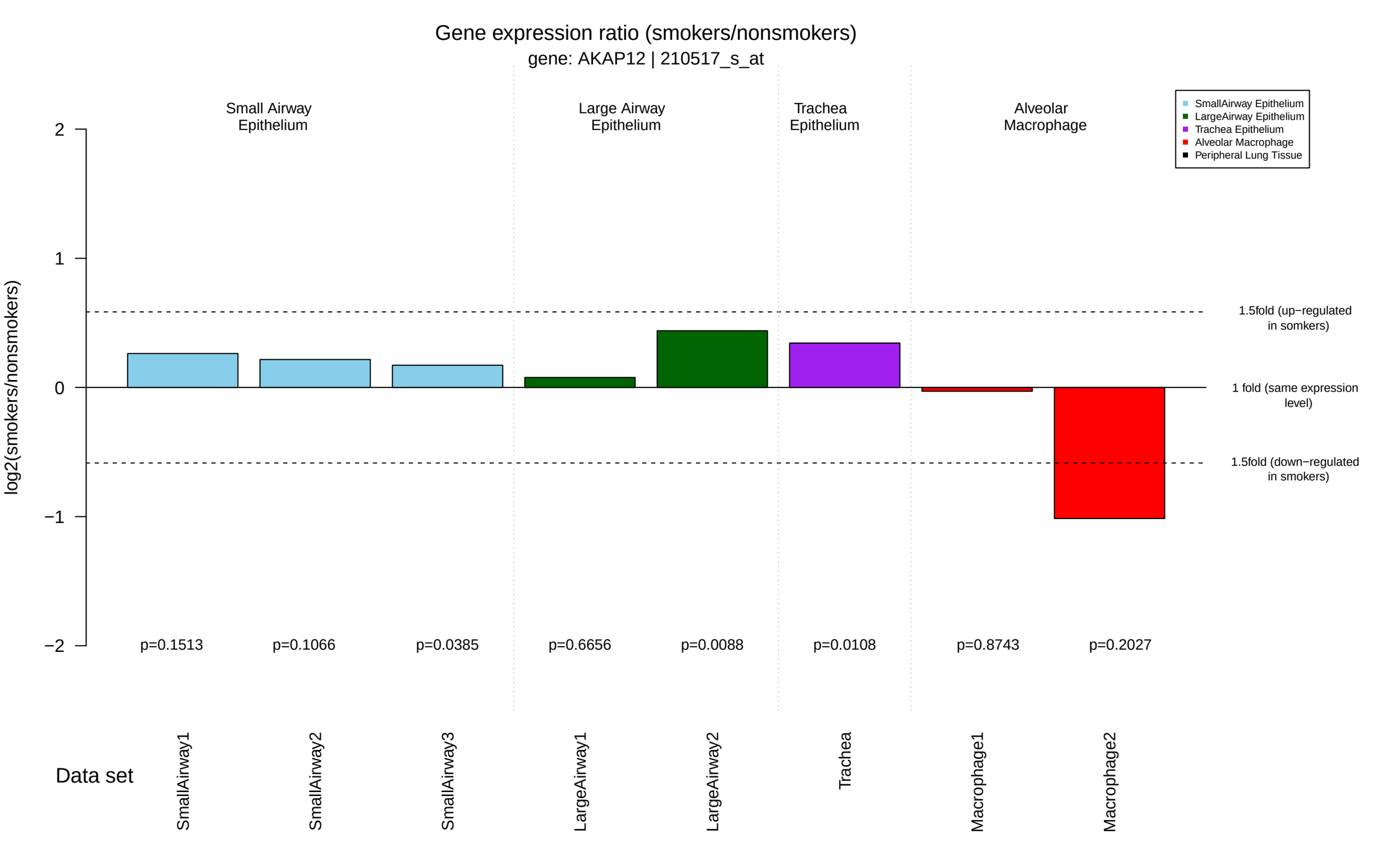
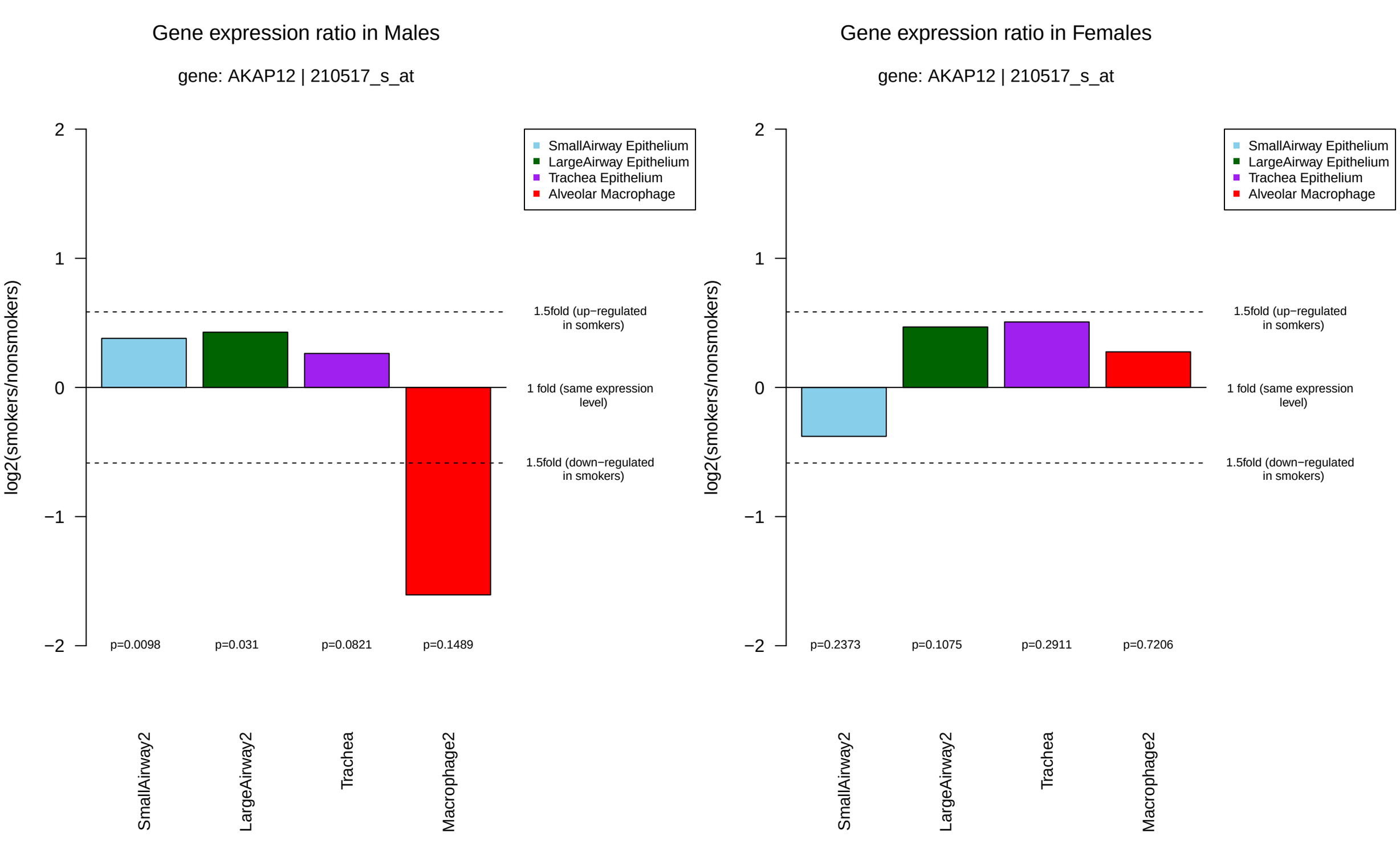
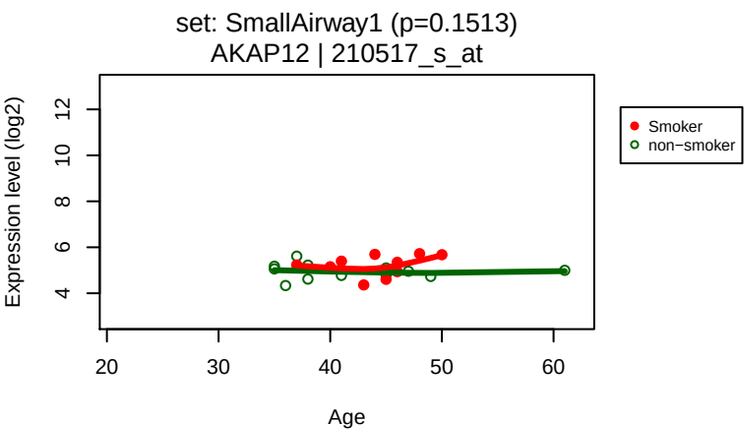
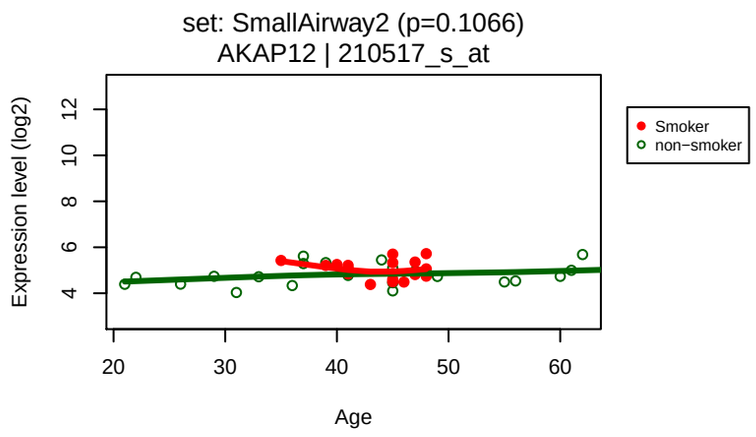
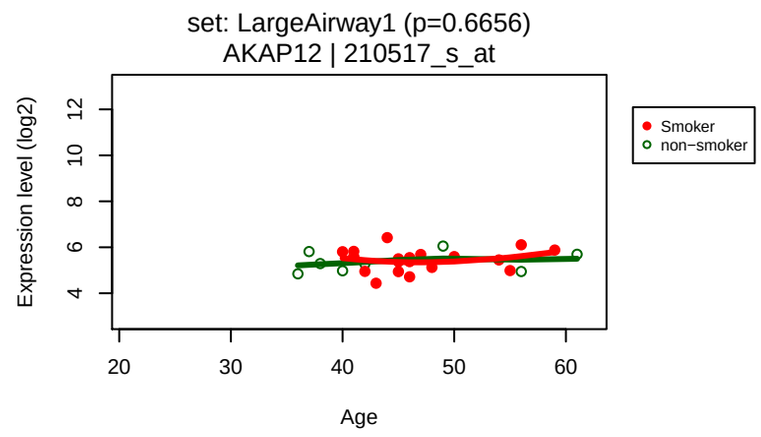

Gene expression of smokers and nonsmokers
gene: AKAP12 | 210517_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1513
p=0.1066
p=0.0385
p=0.6656
p=0.0088
p=0.0108
p=0.8743
p=0.2027
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

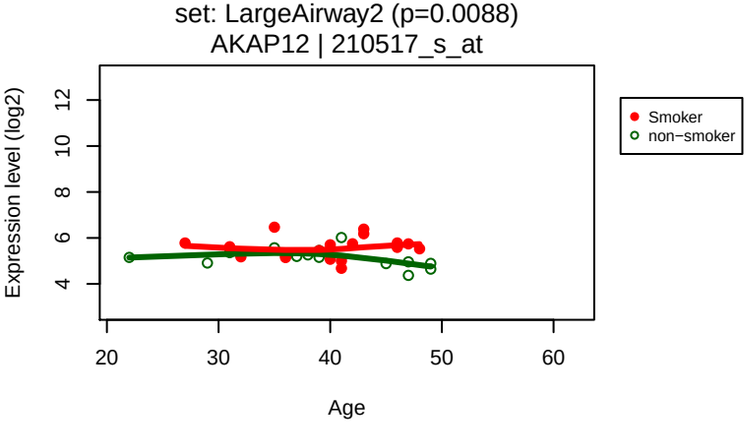

Gene expression ratio (smokers/nonsmokers)
gene: AKAP12 | 210517_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1513
p=0.1066
p=0.0385
p=0.6656
p=0.0088
p=0.0108
p=0.8743
p=0.2027
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

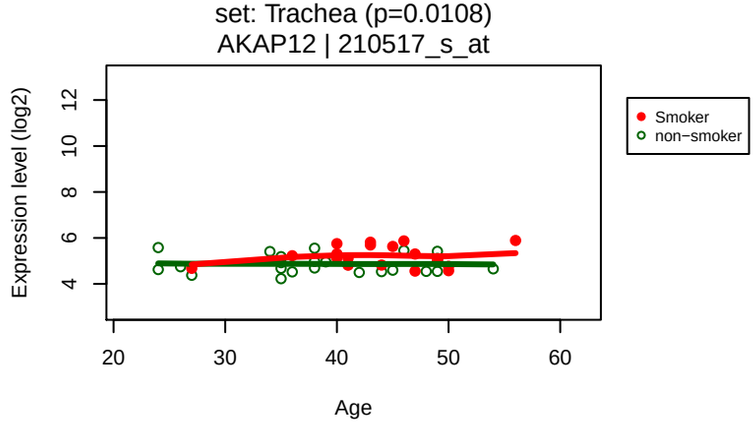

Gene expression ratio in Males
gene: AKAP12 | 210517_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0098
p=0.031
p=0.0821
p=0.1489
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP12 | 210517_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2373
p=0.1075
p=0.2911
p=0.7206
SmallAirway2
LargeAirway2
Trachea
Macrophage2

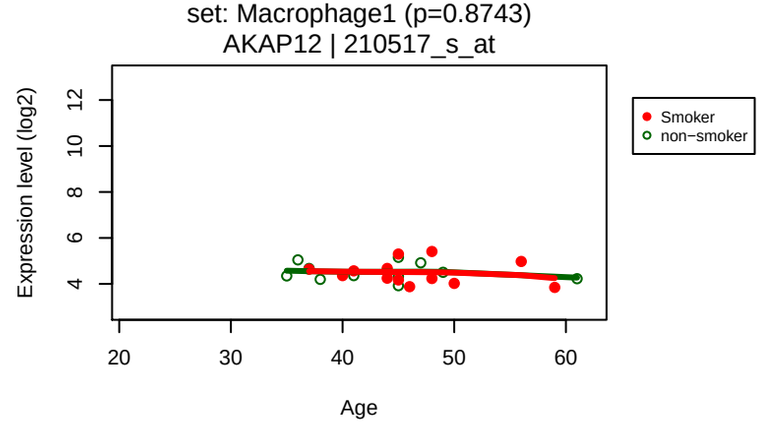

set: SmallAirway1 (p=0.1513)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker

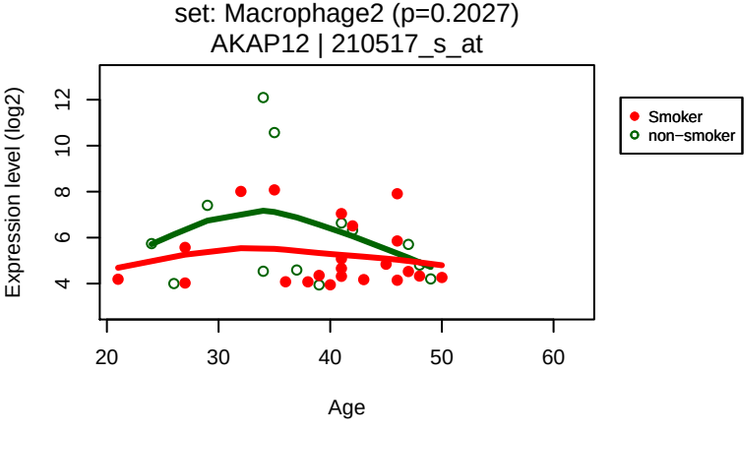

set: SmallAirway2 (p=0.1066)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker

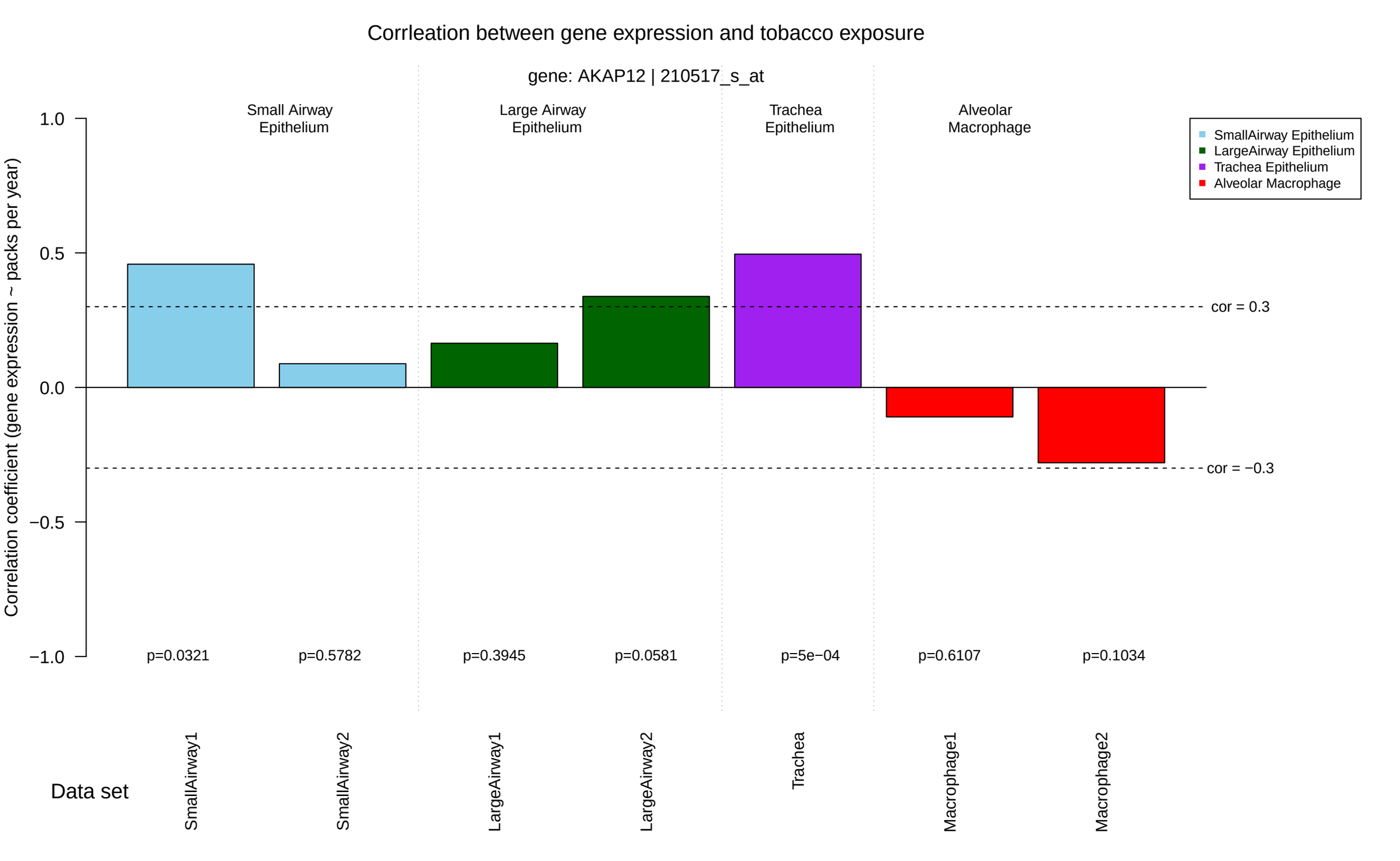

set: LargeAirway1 (p=0.6656)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0088)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0108)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8743)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2027)
AKAP12 | 210517_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP12 | 210517_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0321
p=0.5782
p=0.3945
p=0.0581
p=5e−04
p=0.6107
p=0.1034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2